LIBBY — RECOMMENDATIONS

Recommendations — tnbc-brca1-oligomet-liver-r7p3

Ranked therapeutic options with likelihood, toxicity, mechanism risk, and key references

Generated 2026-05-19

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Profile snapshot

- **Primary site:** breast
- **Histology:** invasive ductal carcinoma, no special type, triple-negative; basal-like by molecular subtyping
- **Stage:** cT2-3 cN1 cM1 — newly diagnosed; right breast primary with ipsilateral axillary nodal involvement and a solitary 1.5cm hepatic lesion consistent with oligometastatic de novo M1 disease (liver-only single-site)
- **ECOG:** 1
- **Age band:** 40-49
- **Sex:** female
- **Biomarkers:** BRCA1 mutated; germline vs somatic not specified; TP53 mutated; PIK3CA mutated (ngs_pending); MYC amplification copy-number amplified; IRS2 amplification copy-number amplified; PTEN copy-number loss; Intrinsic molecular subtype basal-like; TMB 14 mut/Mb; MSI status MSS; Stromal TILs 3+ stromal TILs (TILs-WG criteria, high); ER negative; PR negative; HER2 negative
- **Efficacy/toxicity weight:** 0.7

10 ranked therapeutic options across 6 targetable features. Biomarker workup steps live in the standalone Target validation paths report.

Germline BRCA / HRD-targeting interventions

1. Olaparib 300 mg PO BID (OlympiAD) *(conditional on germline_brca positive)*

Likelihood of effect: High in germline BRCA1 HER2-neg mBC: PFS HR 0.58 (OlympiAD) replicated by HR 0.54 (EMBRACA, talazoparib).

Toxicity burden: Low (G3+ anemia 16%, neutropenia 9%, fatigue, nausea)

Counter-productive MoA:Low (BRCA1 reversion / RAD51 fork-protection rescue is the dominant on-pathway escape (pmid:38302062); baseline ctDNA seeds reversion surveillance.)

Overall: Two independent phase 3 anchors, lowest discontinuation rate on the ranking, oral single-agent — the cat-1 BRCA1 anchor that earns rank 2 in any synthesis that respects guideline fit and evidence replication.****

Key references: PMID 28578601 · PMID 34081848 · NCT02000622

2. Talazoparib 1 mg PO daily (EMBRACA) *(conditional on germline_brca positive)*

Likelihood of effect: High in germline BRCA1 HER2-neg mBC: PFS HR 0.54 (EMBRACA); evidence-quality parity with olaparib, tolerability is the binding differentiator.

Toxicity burden: Moderate (G3+ anemia 39%, neutropenia 21%, thrombocytopenia 15%; dose reductions 53%)

Counter-productive MoA:Low (Same BRCA1 reversion route as olaparib; hematologic compression more often forces dose reduction than off-pathway escape.)

Overall: Evidence-quality parity with olaparib (PFS HR 0.54) but G3+ anemia 39% vs 16% — the contingency if olaparib intolerance emerges, not the parity choice.**

Key references: PMID 30110579 · NCT01945775

Evidence in detail

Olaparib 300 mg PO BID (OlympiAD)

Robson M et al. *N Engl J Med* 2017

- **Disease context:HER2-negative metastatic breast cancer with germline BRCA1/2 mutation (TNBC or HR+; up to 2 prior chemo lines)** (1L-3L) — Germline BRCA1/2 carriers; 50% TNBC, 50% HR+; ECOG 0-1; prior anthracycline and taxane required; up to 2 prior chemotherapy lines for metastatic disease.
- **n:** 302
- **Toxicities:any treatment-related AE** (grade >=3): 36.6% (n=205); **anemia** (grade >=3): 16.1% (n=205); **neutropenia** (grade >=3): 9.3% (n=205); **fatigue** (grade any): 28.8% (n=205); **nausea** (grade any): 58.0% (n=205); **treatment discontinuation due to AE** (grade any): 4.9% (n=205)
- **Efficacy:HR_PFS** 0.58 (95% CI 0.43-0.8); **PFS** 7.0 months; **ORR** 59.9%

Tutt ANJ et al. *N Engl J Med* 2021

- **Disease context:High-risk early-stage HER2-negative breast cancer with germline BRCA1/2 mutation (adjuvant)** (adj) — Germline BRCA1/2 carriers; high-risk early breast cancer (TNBC or HR+); residual disease after neoadjuvant chemo OR pT2/pN1 after adjuvant chemo.
- **n:** 1836
- **Toxicities:any treatment-related AE** (grade >=3): 24.5% (n=911); **anemia** (grade >=3): 8.7% (n=911); **neutropenia** (grade >=3): 4.8% (n=911); **treatment discontinuation due to AE** (grade any): 9.9% (n=911); **MDS/AML** (grade any): 0.2% (n=911)
- **Efficacy:HR_DFS** 0.58 (95% CI 0.46-0.74); **DFS** 3-yr iDFS 85.9%; **HR_OS** 0.68 (95% CI 0.47-0.97)

Talazoparib 1 mg PO daily (EMBRACA)

Litton JK et al. *N Engl J Med* 2018

- **Disease context:Locally advanced or metastatic HER2-negative breast cancer with germline BRCA1/2 mutation** (1L-3L) — Germline BRCA1/2 carriers; 45% TNBC, 55% HR+; ECOG 0-2; prior taxane or anthracycline required unless contraindicated; up to 3 prior chemo lines.
- **n:** 431
- **Toxicities:anemia** (grade >=3): 39.0% (n=286); **neutropenia** (grade >=3): 21.0% (n=286); **thrombocytopenia** (grade >=3): 15.0% (n=286); **fatigue** (grade any): 50.3% (n=286); **treatment discontinuation due to AE** (grade any): 7.7% (n=286)
- **Efficacy:HR_PFS** 0.54 (95% CI 0.41-0.71); **PFS** 8.6 months; **ORR** 62.6%; **HR_OS** 0.76 (95% CI 0.55-1.06)

Pd L1 Cps interventions

1. Pembrolizumab 200 mg q3w + carboplatin AUC2 / gemcitabine 1000 mg/m² d1,8 q21d (KEYNOTE-355)

Likelihood of effect: High in CPS ≥10 1L mTNBC: PFS HR 0.65 (mPFS 9.7 vs 5.6 mo) plus OS HR 0.73 (mOS 23.0 vs 16.1 mo) at final analysis.

Toxicity burden: High (G3+ neutropenia 41%, irAE 5%, treatment-related death 0.4%)

Counter-productive MoA:Low (T-cell exhaustion / B2M-HLA antigen-presentation loss is the canonical primary-ICI-resistance route; rank-1 workup includes baseline B2M/HLA IHC for that surveillance.)

Overall: The defining 1L mTNBC survival regimen at CPS ≥10 (OS HR 0.73, 7-mo median gain); now the alternative cat-1 backbone after SG + pembro took the preferred slot.****

Key references: PMID 33278935 · PMID 35857659 · NCT02819518

2. Sacituzumab govitecan 10 mg/kg d1,8 q21d + pembrolizumab 200 mg q3w (ASCENT-04 / KEYNOTE-D19)

Likelihood of effect: High in CPS ≥10 1L mTNBC: PFS HR 0.65 (mPFS 11.2 vs 7.8 mo); OS still immature.

Toxicity burden: High (G3+ neutropenia ~50%, diarrhea 10%, irAE ~25% any-grade)

Counter-productive MoA:Low (Same T-cell exhaustion / antigen-presentation loss concerns as KN-355; rank-1 B2M/HLA workup applies.)

Overall:NCCN v2.2026 cat-1 preferred for CPS ≥10 over KN-355 on PFS HR 0.65 and DoR 16.5 mo; rank pulled down by single-persona endorsement reflecting fresh elevation and immature OS rather than clinical dissent.****

Key references: NCT05382299 · PMID 33278935

Evidence in detail

Pembrolizumab 200 mg q3w + carboplatin AUC2 / gemcitabine 1000 mg/m² d1,8 q21d (KEYNOTE-355)

Cortes J et al. *Lancet* 2020

- **Disease context:Previously untreated locally recurrent inoperable or metastatic TNBC** (1L) — 1L mTNBC; PD-L1 stratified by 22C3 CPS; ECOG 0-1; primary endpoint in CPS >=10 subgroup.
- **n:** 847
- **Toxicities:any treatment-related AE** (grade >=3): 68.1% (n=562); **neutropenia** (grade >=3): 41.0% (n=562); **immune-mediated AE** (grade >=3): 5.3% (n=562); **treatment-related deaths** (grade 5): 0.4% (n=562)
- **Efficacy:HR_PFS** 0.65 (95% CI 0.49-0.86); **PFS** 9.7 months

Cortes J et al. *N Engl J Med* 2022

- **Disease context:Previously untreated mTNBC, PD-L1 CPS >=10 final OS** (1L) — Same 847-patient cohort; final OS analysis at 44-mo median follow-up.

- **n:** 847
- **Toxicities:any treatment-related AE** (grade >=3): 68.1% (n=562); **treatment-related deaths** (grade 5): 0.4% (n=562)
- **Efficacy:HR_OS** 0.73 (95% CI 0.55-0.95); **OS** 23.0 months

Sacituzumab govitecan 10 mg/kg d1,8 q21d + pembrolizumab 200 mg q3w (ASCENT-04 / KEYNOTE-D19)

Cortes J et al. *Lancet* 2020

- **Disease context:Previously untreated locally recurrent inoperable or metastatic TNBC** (1L) — 1L mTNBC; PD-L1 stratified by 22C3 CPS; ECOG 0-1; primary endpoint in CPS >=10 subgroup.
- **n:** 847
- **Toxicities:any treatment-related AE** (grade >=3): 68.1% (n=562); **neutropenia** (grade >=3): 41.0% (n=562); **immune-mediated AE** (grade >=3): 5.3% (n=562); **treatment-related deaths** (grade 5): 0.4% (n=562)
- **Efficacy:HR_PFS** 0.65 (95% CI 0.49-0.86); **PFS** 9.7 months

Brca1 Mutation interventions

1. Carboplatin AUC2 + gemcitabine 1000 mg/m² d1,8 q21d (PD-L1-agnostic BRCA1-directed backbone)

Likelihood of effect: Moderate to High in BRCA1-mut TNBC: TNT BRCA-mut subgroup ORR 68% vs 33% docetaxel (n=43, p=0.01).

Toxicity burden: Moderate (G3+ neutropenia 30–40%, thrombocytopenia ~15%)

Counter-productive MoA:Low (BRCA1 reversion under platinum pressure mirrors the PARP-i resistance route; serial ctDNA at progression resolves intra-pathway vs off-pathway escape.)

Overall: The PD-L1-agnostic BRCA1-directed backbone; rests on a small TNT subgroup (n=43) plus three decades of carbo/gem experience; sits below SG monotherapy in the post-Feb-2026 NCCN CPS-low stack but pairs more cleanly with later consolidation.****

Key references: PMID 29713086

2. Olaparib 300 mg PO BID + durvalumab 1500 mg q4w as MEDIOLA-style maintenance after platinum + pembrolizumab induction

Likelihood of effect: Moderate in BRCA1 + ICI-favorable phenotype (MEDIOLA ORR 63%, n=34); evidence is single-arm + subset, not RCT-grade.

Toxicity burden: Low (G3+ anemia 12%, neutropenia 9%, pancreatitis 6%)

Counter-productive MoA:Moderate (Critic dissented on the subset-of-subset evidence stack; concensusite dissented on guideline-fit (no registered 1L slot).)

Overall: The chemo-free PARPi + ICI maintenance bet on the canonical-responder phenotype; held back by single-arm evidence stack and guideline-fit gap, retained for tumor-board discussion of post-induction**

maintenance.**

Key references: PMID 32771088 · PMID 31194225 · PMID 41405563 · PMID 38236575

Evidence in detail

Carboplatin AUC2 + gemcitabine 1000 mg/m² d1,8 q21d (PD-L1-agnostic BRCA1-directed backbone)

Tutt A et al. *Nat Med* 2018

- **Disease context:Metastatic or recurrent locally advanced TNBC or BRCA1/2-mutant breast cancer** (1L-2L) — Unselected advanced TNBC (376 pts); 43 patients with germline BRCA1/2 mutations identified by prospective biomarker analysis.
- **n:** 376
- **Toxicities:neutropenia** (grade >=3): 12.0% (n=188); **thrombocytopenia** (grade >=3): 7.0% (n=188); **neuropathy** (grade >=3): 11.0% (n=188)
- **Efficacy:ORR** 31.4%; **ORR** 68%; **ORR** 33%; **HR_PFS** 0.44

Olaparib 300 mg PO BID + durvalumab 1500 mg q4w as MEDIOLA-style maintenance after platinum + pembrolizumab induction

Domchek SM et al. *Lancet Oncol* 2020

- **Disease context:Germline BRCA-mutated HER2-negative metastatic breast cancer** (1L-3L) — Germline BRCA1/2 carriers; HER2-neg mBC; up to 2 prior chemo lines; ECOG 0-1.
- **n:** 34
- **Toxicities:any treatment-related AE** (grade >=3): 32.0% (n=34); **anemia** (grade >=3): 12.0% (n=34); **neutropenia** (grade >=3): 9.0% (n=34); **pancreatitis** (grade >=3): 6.0% (n=34); **treatment discontinuation due to AE** (grade any): (n=34)
- **Efficacy:DCR** 80% (95% CI 64.3-90.9); **ORR** 63%; **PFS** 8.2 months

Vinayak S et al. *JAMA Oncol* 2019

- **Disease context:Advanced or metastatic triple-negative breast cancer** (2L+) — Advanced TNBC; BRCA-mut (n=15) and BRCA-WT (n=27) both eligible; up to 2 prior cytotoxic lines.
- **n:** 55
- **Toxicities:anemia** (grade >=3): 18.0% (n=55); **thrombocytopenia** (grade >=3): 15.0% (n=55); **fatigue** (grade >=3): 7.0% (n=55); **immune-related AE (any)** (grade any): 15.0% (n=55); **immune-related AE** (grade >=3): 4.0% (n=55)
- **Efficacy:ORR** 21%; **ORR** 47%; **ORR** 11%

Rugo HS et al. *Clin Cancer Res* 2026

- **Disease context:Locally recurrent inoperable or metastatic TNBC after induction with 1L pembrolizumab + chemo** (maintenance) — mTNBC with SD/CR/PR on 4-6 cycles of pembro+chemo induction; HRR-mut stratified.
- **n:** 271
- **Toxicities:any treatment-related AE** (grade >=3): 84.4% (n=135); **any treatment-related AE** (grade >=3): 96.2% (n=136); **anemia** (grade >=3): 25.0% (n=135)
- **Efficacy:HR_PFS** 0.98 (95% CI 0.72-1.33); **HR_PFS** 0.7 (95% CI 0.33-1.48)

Tan TJ et al. *Clin Cancer Res* 2024

- **Disease context:Platinum-pretreated advanced TNBC, maintenance after platinum response** (maintenance) — Advanced TNBC with SD/CR/PR on 1L-2L platinum chemo; HRD-positive / BRCA-mut enriched but not required; randomized 1:1.
- **n:** 45
- **Toxicities:nausea** (grade any): 35.0% (n=45); **anemia** (grade any): 31.0% (n=45); **neutropenia** (grade >=3): 4.4% (n=45)
- **Efficacy:PFS_rate** 35.7%; **PFS_rate** 11.8%

Basal Like Subtype interventions

1. Datopotamab deruxtecan ± durvalumab on TROPION-Breast05 (NCT06103864)

Likelihood of effect: Moderate (cross-trial extrapolation, not a within-trial estimate): TROPION-Breast02 PFS HR 0.57 in CPS <10 / ICI-ineligible 1L; Breast05 readout pending.

Toxicity burden: Moderate (G3+ stomatitis 6%, ILD ~3%, ocular events; G3+ TRAE 35%)

Counter-productive MoA:Moderate (Overlapping pulmonary toxicity from Dato-DXd payload + durvalumab pneumonitis; baseline HRCT/PFTs mitigate but no combination safety dossier yet.)

Overall: The trial-preference-anchored 1L slot — randomized against KN-355 with a 50% shot at TROP2-ADC + ICI; effect size still a forward bet on cross-trial extrapolation.****

Key references: NCT06103864 · PMID 40297626 · PMID 33278935

2. Sacituzumab govitecan 10 mg/kg d1,8 q21d monotherapy (ASCENT-03 1L CPS-low / ICI-ineligible)

Likelihood of effect: High in CPS <10 / ICI-ineligible 1L mTNBC (ASCENT-03 NCCN cat-1 elevation); replicated by ASCENT 2L+ OS HR 0.51.

Toxicity burden: High (G3+ neutropenia 51%, diarrhea 10%, febrile neutropenia 6%, treatment-related death 0.4%)

Counter-productive MoA:Low (TROP2-ADC mechanism — payload-driven cytotoxicity; no mechanism-level counter-productive vector specific to BRCA1+ TNBC.)

Overall:NCCN v2.2026 cat-1 preferred 1L for CPS <10 / ICI-ineligible; the CPS-low fallback that displaces carbo/gem in the post-Feb-2026 consensus stack.****

Key references: PMID 33882206 · PMID 38422473 · NCT02574455

Evidence in detail

Datopotamab deruxtecan ± durvalumab on TROPION-Breast05 (NCT06103864)

Cortes J et al. *Lancet* 2020

- **Disease context:Previously untreated locally recurrent inoperable or metastatic TNBC** (1L) — 1L mTNBC; PD-L1 stratified by 22C3 CPS; ECOG 0-1; primary endpoint in CPS >=10 subgroup.
- **n:** 847
- **Toxicities:any treatment-related AE** (grade >=3): 68.1% (n=562); **neutropenia** (grade >=3): 41.0% (n=562); **immune-mediated AE** (grade >=3): 5.3% (n=562); **treatment-related deaths** (grade 5): 0.4% (n=562)
- **Efficacy:HR_PFS** 0.65 (95% CI 0.49-0.86); **PFS** 9.7 months

Sacituzumab govitecan 10 mg/kg d1,8 q21d monotherapy (ASCENT-03 1L CPS-low / ICI-ineligible)

Bardia A et al. *N Engl J Med* 2021

- **Disease context:Pretreated metastatic triple-negative breast cancer (>=2 prior lines including a taxane)** (2L+) — mTNBC after >=2 prior lines in advanced setting, must include a taxane; brain-mets cohort separate.
- **n:** 529
- **Toxicities:neutropenia** (grade >=3): 51.0% (n=258); **diarrhea** (grade >=3): 10.0% (n=258); **anemia** (grade >=3): 8.0% (n=258); **febrile neutropenia** (grade >=3): 6.0% (n=258); **treatment discontinuation due to AE** (grade any): 4.7% (n=258)
- **Efficacy:HR_PFS** 0.41 (95% CI 0.32-0.52); **PFS** 5.6 months; **HR_OS** 0.48 (95% CI 0.38-0.59); **OS** 12.1 months; **ORR** 35%

Bardia A et al. *J Clin Oncol* 2024

- **Disease context:Pretreated metastatic TNBC (final database lock + HER2/TROP2 expression subgroups)** (2L+)
- **n:** 529
- **Toxicities:neutropenia** (grade >=3): 52.0% (n=258); **treatment-related deaths** (grade 5): 0.4% (n=258)
- **Efficacy:HR_OS** 0.51; **OS** 11.8 months; **HR_PFS** 0.41

Oligometastatic State interventions

1. SBRT to the solitary 1.5 cm hepatic lesion as consolidation on MSK SABR protocol (NCT05534438) after documented systemic response

Likelihood of effect: Low to Moderate for OS in breast-specific oligometastatic disease (NRG-BR002 phase IIR PFS HR ~1.31 favors systemic-only); biology-selection argument for this patient is post-hoc.

Toxicity burden: Moderate (SBRT G≥2 AE 29%; 4.5% treatment-related deaths in SABR-COMET)

Counter-productive MoA:Moderate (Treatment-related deaths in SABR-COMET (4.5%) and negative NRG-BR002 phase IIR PFS are the load-bearing mechanism-level concerns against routine off-trial consolidation.)

Overall: The curative-intent layer the preference file was written to enable; honest about NRG-BR002 negative, anchored on NCT05534438 trial enrollment after documented systemic response.****

Key references: NCT05534438 · PMID 33885704 · PMID 34995128 · PMID 30982687

Evidence in detail

SBRT to the solitary 1.5 cm hepatic lesion as consolidation on MSK SABR protocol (NCT05534438) after documented systemic response

Chmura SJ et al. *JAMA Oncol* 2021

- **Disease context:Newly oligometastatic breast cancer (1-4 metastases) of any subtype, on standard systemic therapy** (1L+ consolidation) — Breast-specific cohort; 1-4 oligometastases; on standard systemic therapy; randomized to add consolidation vs continue systemic alone.
- **n:** 125
- **Toxicities:any treatment-related AE** (grade >=3): 12.0% (n=65)
- **Efficacy:HR_PFS** 1.31; **PFS_rate** no difference at 3 yr

Khan SA et al. *J Clin Oncol* 2022

- **Disease context:De novo stage IV breast cancer with intact primary tumor, no progression after 4-8 mo of optimal systemic therapy** (1L+ consolidation) — Stage IV intact primary; 50% HR+/HER2-, 29% HER2+, 10% TNBC, 11% other; bone-only 31%, viscera-only 26%, both 27%; median age 55.
- **n:** 256
- **Toxicities:surgical complications** (grade any): 15.0% (n=125)
- **Efficacy:HR_OS** 1.11 (95% CI 0.82-1.52); **OS_rate** 68.4%; **OS_rate** 67.9%

Palma DA et al. *Lancet* 2019

- **Disease context:Controlled primary solid tumor with 1-5 metastatic lesions (mixed histologies, including breast)** (any) — Multiple primary sites; 18% breast; 1-5 oligometastases; controlled primary.
- **n:** 99
- **Toxicities:any treatment-related AE** (grade >=2): 29.0% (n=66); **treatment-related deaths** (grade 5): (n=66)
- **Efficacy:HR_OS** 0.57 (95% CI 0.3-1.1); **OS** 41 months

Myc Amplification interventions

1. ZEN-3694 (BET inhibitor) + pembrolizumab + nab-paclitaxel on NCT05422794

Likelihood of effect: Low for characterized efficacy (phase 1b dose-finding, no readout); class extrapolation from MYC-amp / BET-i mechanism only.

Toxicity burden: High (BET-i thrombocytopenia 20-30%, nab-paclitaxel neutropenia 30%, pembrolizumab irAE — uncharacterized triplet)

Counter-productive MoA:High (Conservative veto on toxicity-as-mechanism grounds: three independent myelosuppressive mechanisms with no characterized triplet AE algorithm.)

Overall: Conservative veto on cumulative toxicity without a triplet safety dossier; mechanism is aligned with the patient's MYC-amp / basal / TIL-rich axis but evidence is too thin to act on.****

Key references: NCT05422794 · PMID 26406987